

A NEW APPROACH TO TEACHING FUNDAMENTAL STATISTICAL CONCEPTS AND AN EVALUATION OF ITS APPLICATION AT UCT

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Summary: This paper starts by considering the standard theory-based or menu driven approach to the teaching of Statistics and argues that, although the approach makes the subject easy to teach and easy to learn, it, more insidiously, also provides a mechanistic framework for students to pass who, in reality, have very little in-depth knowledge about the subject. The paper reviews the record of initiatives in the teaching of Statistics which have used computer technology to best communicate statistical ideas. It is suggested in the paper that the simulation teaching model may be particularly useful in facilitating the more effective teaching of Statistics and we illustrate how such a model within the Microsoft Excel spreadsheet environment can be a very effective teaching tool. This program has built-in programming functionality known as Visual Basic for Applications (VBA) which is ideal for the writing of simulation-based teaching tools in Statistics.

At the University of Cape Town (UCT) we have been examining the didactic efficacy and appropriateness of this method in a large (1100 students) first year Statistics course. The paper reviews the results of the surveys conducted and indicates that this innovative approach is both popular with students and very effective at putting over fundamental statistical ideas to the full spectrum of student capability. We note that university teaching in South Africa is itself in a state of flux as it grapples with a very different profile of student from that of twenty years ago. The paper concludes that this initiative constitutes a very necessary effort to modernise and improve the teaching of Statistics to the South African student body, in particular, but also that we see the method

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as one which has general relevance to both first world and third world teaching of the subject of Statistics.

1. Introduction and contextualisation

Traditional methods of teaching statistics, as well as maths, can be seen as flawed in several ways. Generally, complex statistical ideas are taught from a theory-based standpoint using a chalk-and-board written-down-and-copy approach. The focus of the teaching model is often to impose on students large chunks of theoretically based material with its associated technicalities. In some cases, students must memorize the complicated proofs contained in this material and then reproduce them in an examination. Moreover, there is often no attempt to examine whether any understanding extends beyond the limits of these memorized proofs. Probing below the surface of this taught material often exposes a very fragile grasp, and often no grasp at all, of the fundamental concepts underlying such material. Of course, not all statistics courses and particularly those first courses catering for non-mathematically orientated students require the memorizing of proofs; however, even when statistics is taught as a so-called “service” course with minimal mathematical foundation, teachers can often lapse into a menu driven approach to the subject with little focus on putting across a real understanding of the fundamental concepts underpinning the subject.

In some cases, particularly when very bright students are taught by such methods, the results are very encouraging and often such students can be shown to have a strong and deep grasp of the material, although of course it can be argued that such students would have a very deep grasp of material however it were taught, or even if it were not taught at all (the student simply being

given the textbook). Hence the performance of the very best students is often not central to a debate on teaching methodology. Of considerably greater importance is the median student and in fact all students apart from the very best. We will argue in this paper that the theory-based or menu driven approach to the teaching of Statistics outlined above can be seen as insidious and elitist. Insidious because it has a certain appeal to both teachers and students. It is easy to teach and easy to learn. It thus represents a mechanistic framework for students to pass who, in reality, know very little. Only the very bright have anything except the most superficial grasp of the material. By, in reality, catering primarily to the very bright, it is a method that is also elitist in a society promoting proper access to proper learning methods for all.

Up to now the didactical response to dealing with students who were increasingly ill prepared for the first year statistics course, had been to simplify the material to the point where it has become more focused on perfecting the application of recipes than on facilitating a deep understanding of the statistical fundamentals. This, if anything, has exacerbated the current problem.

The University of Cape Town has a first year Statistics course aimed primarily at Commerce students who, in general, go on to do only one further course in Statistics (at the second year level) and Science students who are not majoring in Statistics. This new approach to teaching introductory statistics was implemented in this first year course which comprises some 1100 students, the majority of whom are from the Commerce Faculty. The teaching team comprised seven lecturers with varying degrees of teaching experience. The cohort of students represented a useful group with which to analyse the effectiveness of this method and in the latter part of this paper we present results from a survey we conducted on the group which attempted to determine the receptiveness of the students to the new approach and to what extent the key

ideas we tried to put across were absorbed by the students. We also document the reaction of the students and the teaching staff to the introduction of this new approach.

2. Using Excel and VBA as a teaching vehicle

2.1 Formulating a new approach to teaching Statistics

It has been widely recognised that traditional teaching methods for courses in statistics can be improved and that in their current guise, they are often not successful in their aims. Major initiatives in this regard include the Consortium for the Advancement of Undergraduate Statistics Education (CAUSE) initiative in the USA, a grouping which, inter alia, encourages the development and dissemination of undergraduate teaching resources in the field of Statistics and the promotion of Statistics education research as a recognized discipline with a visible presence. In addition, the American Statistical Association funded Guidelines for Assessment and Instruction in Statistics Education (GAISE). This group has similar goals to CAUSE, but focuses on both introductory college courses and so-called “K-12”, education, that is the sum of primary and secondary education. Dedicated Teaching-of-Statistics conferences, such as the two recent IASE Satellite Conferences on Statistics Education and the Communication of Statistics in Sydney (2005) and in Guimarães (2007) have brought considerable focus to the issues surrounding the teaching of statistics.

A major thrust of these current initiatives has been the encouragement of the use of computer technology to best communicate statistical ideas. Moreover, the ideas are often best communicated using graphical techniques and, where possible, within the framework of interest of the audience. It is suggested in this paper that the simulation teaching model may be a very effective tool in facilitating current approaches to the more effective

teaching of Statistics. In particular, this simulation teaching model works well within the Microsoft Excel spreadsheet environment. This program has built-in programming functionality, known as Visual Basic for Applications (VBA), which is ideal for the writing of simulation-based teaching tools in Statistics. While many statistical teaching modules have been developed within the Excel environment (such as Discovering Important Statistical Concepts Using Spreadsheets (DISCUS) developed in the Faculty of Engineering and Computing, Coventry University, the authors are unaware of the use of a suite of VBA developed modules for statistical teaching purposes.

A comprehensive suite of modules has been developed by us at the first year statistics level which uses this simulation approach to teach basic statistical ideas using repeated sampling methods through simulation via a set of modules written in VBA. This suite of programs has been designed to facilitate the learning of statistics in a contemporary didactic setting for a comprehensive range of student talents, ranging from the very mathematically gifted to those who are required to complete a Statistics course as a minor part of their curriculum.

We believe that core to the understanding of many statistical concepts is the idea of repeated sampling and that without confronting this notion practically, many of the basic concepts in statistics remain elusive to the student. The approach currently being piloted in our first year course is rather to reduce the scope of the material covered and to use simulation tools as a medium of teaching. The purpose of simulation exercises based on Excel is to make the statistical concepts visual (i.e. happen before the students' eyes) and open to probing with "what if" type questions. The aim is to produce students who, at the end of the introductory year to statistics, have a clear understanding of basic statistical concepts, translated into an ability to interpret statistical analyses and

tests, and appropriately manipulate data. In addition, the code is made freely available to the students; thus curious and creative students may use it as a vehicle to enhancing nascent computer programming skills.

Moreover, such an approach also develops skills in Excel, a programme that has become the *de facto* “pencil and paper” desktop package right across the commercial sector and hence exposure to this package at any level is an added plus for students. In addition, there are no copyright problems with the package at a South African university since Microsoft (SA) has offered Microsoft Office (which includes Excel) free of charge to the SA education community.

2.2 Previous work on teaching statistics with classroom computer software

The idea of using simulation to explain statistical concepts has been around for some time; the earliest published work being that of Shulte (1979), but has increased in popularity as personal computers have increased in power and functionality. Mills (2002) has provided a useful overview of using simulation methods in the teaching of Statistics. He cites the earlier frequent use of Minitab, for example, in Shibli (1990), Hubbard (1992) and Franklin (1992) for generating simulations.

Most authors have focused on the application of the simulation teaching method to a particular teaching example; for example, Shibli (1990) considered the binomial distribution and Franklin (1992) the case of linear regression. The most comprehensive set of examples developed has been by Chance and Rossman (2006) who have presented a number of lessons in Statistics using Minitab macros and Java applets, but as a general presentation of the simulation approach for teaching Statistics, rather than, as in this paper, a comprehensive set of support tools for a full course. These authors have not

presented any data on the effectiveness of their technique. More recently, there have been some attempts at quantifying the efficacy of a simulation approach in teaching statistics; for example, Parvate et al. (2007) has used the *Fathom* program developed by William Fivaz to assess the teaching of different measures of dispersion to a small class (16 members) of students. In addition, Francis et al. (2007) has used a sample of 76 students to test the effectiveness of a computer based simulation to enhance the developmental learning of the key concepts underlying statistical inference at the Swinburne University of Technology in Australia.

Mills (2002) does not, however, cite the use of VBA (embedded in Excel) for simulation generation in a teaching context. In conjunction with VBA, Excel provides a full set of powerful programming tools as well as the totality of Excel's numerical and graphical capability. In addition, in South Africa, Excel is entrenched in the commercial sector as a *de facto* standard and since the preponderance of our students are employed in this community we were encouraged to use VBA in MS Excel. Clearly the opensource programming languages R and Java (core code has been open source since May 2007) are both rich languages and environments for statistical computing and graphical analysis and either could have been used as foundation languages for this exercise. However, few of our students will work in environments which use these languages to the extent which Excel is used. The simulation teaching modules discussed here were thus all developed using VBA.

3. The Fundamental statistical concepts that we focus upon

As outlined above, we feel that traditional methods of teaching statistics have failed a large proportion of students, particularly in the era when barriers to

tertiary education have been removed and imperatives to extend educational opportunities are high. This has manifested itself in very low levels of statistical literacy, even in, for example, those students entering our Honours programme, despite high grades and apparent abilities to perform statistical manipulations.

The key statistical concepts that are often elusive in a traditional-syllabus introductory course in statistics include:

- Random variation
- Distributions (empirical vs. theoretical¹)
- The sample mean as a random variable
- The distribution of the sampling mean
- The effect of sample size
- The purpose of repeated sampling/ simulation
- The Central Limit Theorem
- Statistical significance
- Statistical power
- Testing hypotheses about the regression coefficients

The approach of using simulation to teach statistics is implemented through VBA programs which allow interactive changes of parameters and which demonstrate the above listed elusive concepts. The new approach can transform the teaching of statistics as a dull and fearsome subject to that which presents a fascinating discipline which has the power to reveal the mysteries of chance, risk and uncertainty, which underline our human existence.

We should point out that this type of teaching methodology represents no panacea. Its value is in helping students *understand* fundamental statistical concepts. It cannot help to replace the rich experience necessary for someone

¹ By theoretical we infer here the distribution of outcomes that are expected under the assumptions made in the simulation; that is, the formulaic distribution under consideration.

to properly apply and implement the wide ranging set of available statistical tools necessary to reach valid conclusions; see, for example, Lipson (2007).

4. Examples of how the “simulation approach” works

A number of modules have been written to show how statistical distributions can occur naturally in situations that can occur in real life. By way of example, we will consider below the module written to demonstrate the natural occurrence of data having a binomial distribution. Note that each programme is driven by underlying VBA code such that, by clicking the appropriate button, VBA code which is embedded in the Excel sheet can be executed.

The programmes have certain common themes. They simulate situations according to repetitive experiments and generate distributions of key random variables. Each programme allows one to vary the number of experiments or simulations. Clearly the larger the number of experiments or simulations, the closer the generated empirical distribution of the random variable of interest will (on average) get to the theoretical distribution.

In the example below, we consider 5 mice working their way through a very simple maze. There is only one turn in the maze with a large sign that says “Turn Right”. If the mouse turns right, the mouse is said to have solved the maze. If the mouse turns left, the mouse has not solved the maze. Each experiment comprises recording the number of mice (out of 5) that correctly solve the maze; these are termed the successful mice. The aim of this experiment is to consider scenarios relating to the intelligence of mice and their ability to read the sign.

Note that this application of a binomial distribution was set up to cater for the realm of experience present amongst our first year students. An alternative

simulation for an introductory course in Statistics for those entering our Maths of Finance stream could look at testing the Random Walk (RW) model of share price behaviour. This would consider some predetermined set of shares and simulate how many of them moved up and how many moved down over some fixed period under various incarnations of the RW model.

In these contexts, the simulation programme acts as both a lead-in to the binomial distribution as well as stimulating students to think about hypothesis testing. In the mouse example, students should consider the issue of how many mice out of the 5 on average would be expected to solve the maze if mice had some measurable intelligence or alternatively no intelligence.

Parameters in red are able to be set on the sheet. In this example we first consider 20 experiments where we specify the parameter p , the probability of the mouse turning right as 0.5 (50%).

If we click on the “Run the experiment” button, we will obtain a random set of simulation results based on a p equal to 0.5. The number of successes over these 20 experiments, or trials, ranges from 0 through to 4; thus note that in this particular run there were no cases when 5 out of 5 mice were successful.

4.1 Sheet for binomial demonstration: # experiments equals 20:

BINOMIAL		Results for the 5 mice finding there way out the maze						#Successes	Frequency
©GDIB - UCT - 20 06								0	1
		If Mouse solves maze = 1						1	3
		If Mouse fails to solve maze = 0						2	6
# Experiments	Experiment #	Mouse #1	Mouse #2	Mouse #3	Mouse #4	Mouse #5	Total Successful Mice	3	5
20	1	0	0	0	1	1	2	4	5
#Mice (Fixed)	2	0	1	0	0	0	1	5	0
5	3	1	1	0	0	1	3	Total	20
P(Mouse solves maze)	4	0	0	1	0	1	2	Compare with the Theoretical distribution	
0.5	5	0	0	1	0	1	2		
	6	1	0	0	1	1	3		
	7	0	0	0	0	0	0		
	8	1	0	0	1	0	2		
	9	1	0	0	1	0	2		
	10	1	1	0	1	1	4		
	11	1	1	1	0	0	3		
	12	1	1	1	1	0	4		
	13	1	1	0	1	0	3		
	14	1	1	0	1	1	4		
	15	0	1	0	1	1	3		
	16	0	0	0	1	1	2		
	17	0	0	0	0	1	1		
	18	0	0	1	0	0	1		
	19	0	1	1	1	1	4		
	20	1	1	0	1	1	4		

If we now click on the “Plot the histogram” button, we will see an empirical distribution of the 20 trials. The following graph will pop up.

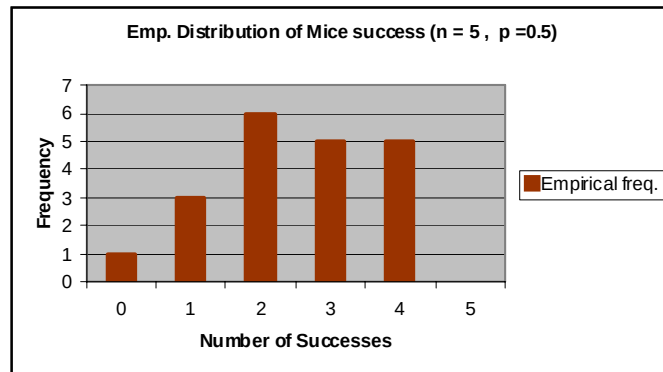


Figure 1. A Simulated (Empirical) distribution (# experiments = 20)

We might then want to explain to students that this is simply one empirical distribution (based on the particular results we randomly generated). If we re-run the experiments we will get a different empirical distribution. However, there will be a relationship between the theoretical binomial distribution and the empirical distribution and we may want to compare the two. If we click on the “Compare with theoretical distribution” we would obtain the following.

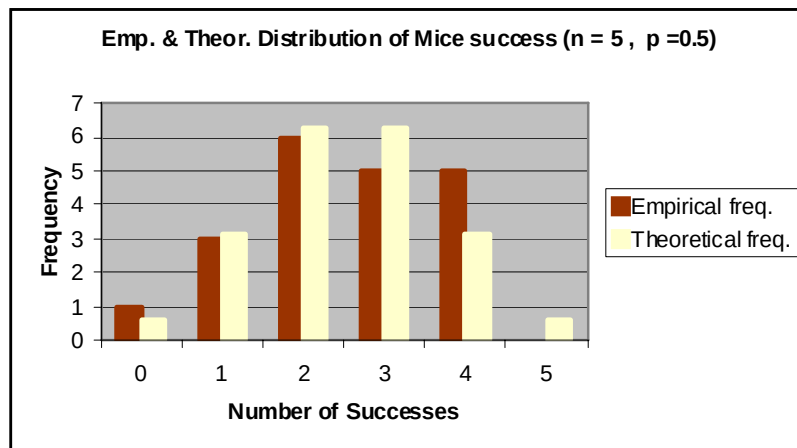


Figure 2. Comparing an empirical with the theoretical distribution (# experiments = 20)

We could now increase the number of experiments to 200. This would result in the sheet below; note that experiments 21 through to 200 are not listed, but the summary results for the number of successes are listed.

4.2 Sheet for binomial demonstration (# experiments = 200)

BINOMIAL		Results for the 5 mice finding there way out the maze						#Successes	Frequency
©GDIB - UCT - 20 06								0	4
		Run the experiment						1	40
		Plot the histogram						2	49
								3	55
								4	43
# Experiments	Experiment #	Mouse #1	Mouse #2	Mouse #3	Mouse #4	Mouse #5	Total successful Mice	5	9
200	1	0	0	0	1	1	2	Total	200
#Mice (Fixed)	2	0	1	0	0	0	1	Compare with the Theoretical distribution	
5	3	1	1	0	0	1	3		
P(Mouse solves maze)	4	0	0	1	0	1	2		
0.5	5	0	0	1	0	1	2		
	6	1	0	0	1	1	3		
	7	0	0	0	0	0	0		
	8	1	0	0	1	0	2		
	9	1	0	0	1	0	2		
	10	1	1	0	1	1	4		
	11	1	1	1	0	0	3		
	12	1	1	1	1	0	4		
	13	1	1	0	1	0	3		
	14	1	1	0	1	1	4		
	15	0	1	0	1	1	3		
	16	0	0	0	1	1	2		
	17	0	0	0	0	1	1		
	18	0	0	1	0	0	1		
	19	0	1	1	1	1	4		
	20	1	1	0	1	1	4		

Similarly, we show the empirical distribution and a comparison between the empirical and the theoretical distribution (under an assumption that $p = 0.5$) when the number of experiments is increased to 200.

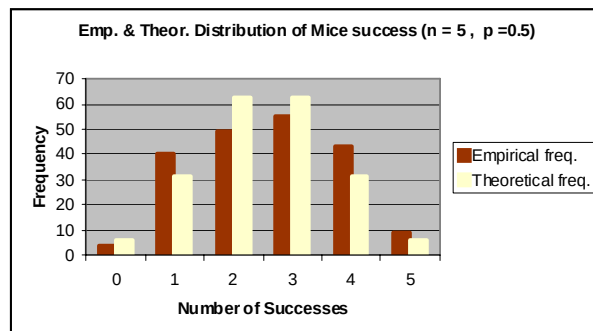


Figure 3. Comparing an Empirical with the theoretical distribution (# experiments = 200)

4.3 Contrasting simulation with the traditional teaching approach

The traditional approach is to teach the binomial distribution through a technical formulaic approach. For the above example, this would comprise deriving the mathematical formula for the binomial distribution and computing the probabilities that 0, 1, 2, 3, 4 or 5 mice (out of 5) made it though the maze under the assumption that $p = 0.5$. From these probabilities, students would compute theoretical frequencies for each of the above events given some fixed number of experiments.

This approach, however, gives students no sense of the statistical distribution of successful mice obtained through repeated sampling (simulation). Through the process of repeatedly generating random outcomes, students can, in contrast to the traditional approach, capture the notion of an empirical distribution which itself varies randomly. They can then, in particular, demonstrate that as the number of experiments increases, so the empirical distribution converges onto the theoretical distribution.

A concept which students often grapple with, and to which a lack of adequate explanation may undermine their interest in the subject, is the difference between expected outcomes and actual outcomes, particularly when the sample size is small (typically one for the individual in a real-life situation!). Simulation allows us to demonstrate that unusual events DO happen but that they can be shown to be part of a pattern that is underpinned by probability and randomness, and not some mystical event which belies these concepts.

To some extent this may have something to do with the fact that most students have had little exposure within their mathematical education to consideration of outcomes which follow a statistical distribution.

Mathematical training focuses almost exclusively on deterministic evaluations; that is in calculating one answer to a particular problem. The whole idea of a distribution of outcomes is something outside the realm of most students' education and requires a shift of focus and concept. The simulation approach accommodates and facilitates this conceptual move for the student.

Moreover, wherever possible, the simulation examples try to support the learning process with examples that are directly from the student's realm of experience; hence the example of mice working through a maze and the distribution of outcomes under various assumptions about mouse intelligence; a setting which is easily understandable and which can easily be demonstrated. This simulation then naturally leads on to one that considers statistical hypothesis testing. That is, how would one use the results of such an experiment to directly test the level of the mice's skill in navigating the maze? The hypothesis testing simulation can then carefully examine the probability of Type 1 and Type 2 errors; that is, considering the probability of falsely concluding that they do have intelligence when they do not, as well as the probability of concluding that mice have no intelligence under the range of scenarios where they possess varying forms of intelligence.

5. The experience of teaching with simulation tools

A basic aim of the new teaching thrust was to produce students who had a clear understanding of basic statistical concepts which would then translate into an ability to interpret statistical tests and appropriately manipulate data. The key concepts that were thought to be elusive to students encountering statistics for the first time were outlined in section 3. The following suite of independent VBA simulation modules was then used as teaching support for the teaching program: Uniform distribution (random sampling);

Binomial distribution; Poisson distribution; Normal distribution (Central Limit Theorem); Hypothesis testing of sample mean; Hypothesis testing of simple linear regression coefficients.

Weekly co-ordination meetings were held to ensure that the new teaching approach was consistently applied. This proved to be quite a challenge and not all staff were equally enthusiastic about adopting a new teaching approach. Any change in teaching style or material requires substantial extra effort on the part of the teachers and it also required the teaching team to take a “leap of faith” in abandoning previously familiar lecture notes in favour of a fresh approach to familiar material. The teaching team were asked to “test drive” the programs prior to using them in lectures and many cosmetic changes were made to adapt the programs to the pace and level of first year students. The team also tried to make the exercises fun to use as well as humorous.

Each of the VBA programs was used to illustrate a fundamental statistical concept within an environment in which the student would develop an appreciation for the effects of random variation on the sample. The teaching team came to hold the strong view that the insights gained using this simulation approach could never be attained using traditional “static” data set illustrations. Importantly, it was felt that the simulations provided a space in which students could ask “what if” questions and debate material which is typically presented to first year students as incontrovertible and consequently not likely to stimulate debate.

5.1 Teachers’ experiences of using the simulation approach

Teachers who participated in this project to transform first year teaching using the simulation approach described their experiences as follows (we will not name the teachers, simply label them as $T_1, T_2, \dots, T_8$).

5.1.1 T_1 (18 years of statistics teaching experience): Cited several advantages, including that it provided a more interactive mode of teaching which held students' attention more readily. Students who had been led to believe that stats was about exciting, even magical, things such as gambling and forecasting and predicting extreme events, often were seen to "dull-out" with the mathematical formula approach and lose interest. The new approach went some way to keeping the "statistical light" alive. There was a need to constantly relate material to real world events in the realm of students' experience, so they could more easily relate to the work covered.

5.1.2 T_2, T_3, T_4 : (5/4/4 years of statistics teaching experience): said that using the simulation approach had invigorated the teaching experience in first year, particularly compared to previous years. There was a "chalk and cheese" difference with previous years – students now were much more interested and lively. Raised enthusiasm levels meant that students were easier to teach in general.

5.1.3 T_5 : (Tutor and graduate student): spoke about how useful he found the simulation approach when an Honours student and how he had experienced it as a refreshing and positive didactic break from the standard statistical teaching he had received. He also mentioned how much he had learned from his Honours project on statistical power (using simulation in VBA). As a tutor he related how students could be taught a mathematical approach but often had a shallow idea of what they were doing.

5.1.4 T_6 : (8 years teaching experience): used the approach to recap first year statistical concepts with students at the beginning of their second year. She found that students' questions and responses to the simulation outputs indicated a profound lack of understanding of basic concepts, even apparent in those who had achieved high marks at the end of first year with the standard

mathematical approach. The importance of using real-life data sets (in addition to theoretically generated data sets) was also emphasised.

5.1.5 T_7 : (20 years teaching experience): was very supportive of the approach and reported positive teaching experiences. Commenting on possible resistance to the work involved in adopting a new approach, his view was that it was worth it and that he had experienced added student enthusiasm from using this approach.

5.1.6 T_8 : (2 years of teaching experience) reported positive teaching experience with the new approach and saw it as an important way forward with no disadvantages.

Despite the fact that these teaching changes were introduced with a team which included a number of relatively inexperienced teachers, as well as tutorial material which was substantially in need of overhaul, the results were found to be very positive. On the basis of this initiative, a decision was taken to extend the simulation approach to other courses in the department.

6. Evaluating the learning experience

It is difficult to thoroughly evaluate the effectiveness of the new approach as there is essentially no baseline point of comparison to test the belief that students taught statistical concepts (such as random variation; classical hypothesis testing and sampling distributions) through conventional approaches may, in many cases, have a very poor grasp of these concepts. However, in an attempt to probe the levels of understanding attained through this method of teaching, an evaluation questionnaire (given below in the Appendix) was developed. This questionnaire asked students for their experience of simulation as a teaching tool, and also attempted to probe their understanding of some of the 'problematic' concepts listed in section three. All

first year students exposed to the new approach were given the questionnaire to complete during class time. The summary results are shown below.

6.1 Questionnaire results

The number of questionnaires completed and captured (773) relative to students registered (1133) is probably an accurate reflection of the proportion of students who attend lectures. Due to the fact that questionnaires were not completed under exam conditions and that students were at liberty to consult with neighbours whilst completing their forms, a high level of “copying” was noted.

The following pie-charts reflect the questionnaire responses to Questions 1 through to 5, which comprised broad questions attempting to establish whether the overall course was interesting and enjoyable and whether the key concepts of theoretical (probability) distribution and empirical distribution had been understood.

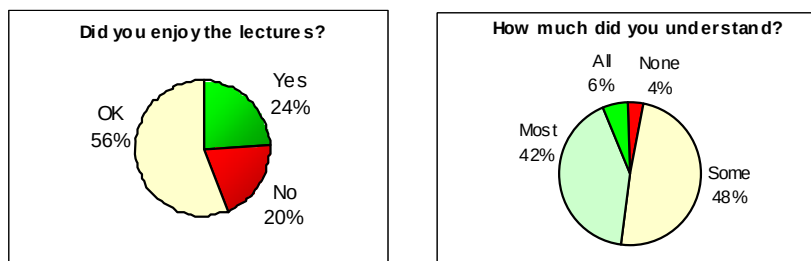


Figure P.1. Enjoyment of lectures **Figure P.2.** Perception of own understanding

Clearly the students have a higher perception of their own levels of understanding (Figure P.2) than their responses to questions of an explanatory style indicate (Figures P.3, P.4 and P.5).

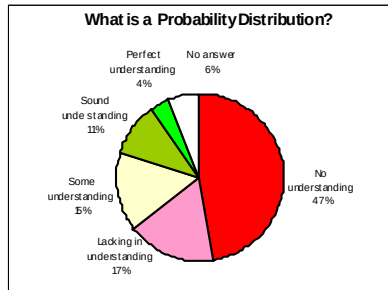


Figure P.3. Understanding of
"probability distribution"

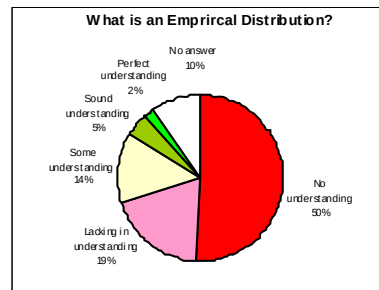


Figure P.4. Understanding of
"empirical distribution"

Evaluating the degree to which students understand sophisticated statistical concepts is a task which requires some subjective assessment, bearing in mind that the concepts are generally multi-faceted and the students may or may not have the linguistic skills to express their knowledge. We also expressly did not want to test the students' ability to regurgitate a standard "dictionary" definition of the terms. For this reason the students were asked to explain the concepts in their own words.

The students' answers to Q3 and Q4 were subjectively assessed by members of the teaching team. Table 1 and Table 2 below show, by way of examples, the manner in which the students' answers were assessed.

Table 1. Marking scheme for Q3: *What is a probability distribution?*

No understanding	No correct elements in answer E.g. “The distributions cannot be predetermined.”; or “I really don’t know”
Lacking in understanding	Some correct elements in answer but substantially misguided E.g. “the different probabilities that a function will take on certain values”
Some understanding	Some correct elements in answer and “on the right track” E.g. “The likelihood of a certain random variable occurring can be determined using a distribution of probabilities”
Sound understanding	Most elements of the answer are correct E.g. “A distribution that illustrates the various probabilities of certain events.”
Perfect understanding	All elements of answer are correct and present E.g. “A function that gives the probabilities that a variable equals each of a sequence of possible values e.g. binomial, exponential, normal and Poisson distributions

Table 2. Marking scheme for Q4: *What is an empirical distribution?*

No understanding	No correct elements in answer E.g. “A concept in statistics”; “A continuous probability distribution”; “A cumulative distribution function”
Lacking in understanding	Some correct elements in answer but substantially misguided E.g. “A calculation where the occurrence of events in an experiment is calculated manually, e.g. counted.”
Some understanding	Some correct elements in answer and “on the right track” E.g. “A distribution compiled by using observed data from a data set.”
Sound understanding	Most elements of the answer are correct E.g. “A distribution of probabilities which illustrates the true results of data – i.e. it is not theoretical as it is with a probability distribution.”
Perfect understanding	All elements of answer are correct and present E.g. “a p.m. for p.d.f. that we observe when we take a finite random sample. As sample size increases it becomes more representative of the population.”

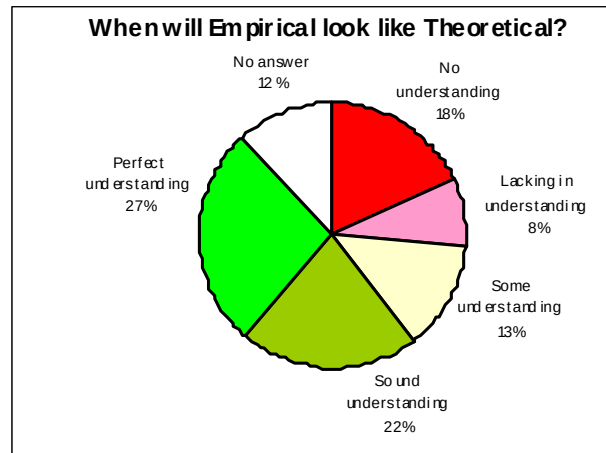


Figure P.5. When will empirical and theoretical distributions be the same?

Question 5 highlighted some fundamental conceptual errors:

- (i) Many students were unable to make the distinction between sample size, number of repetitions of an experiment and sample space, claiming a need for “infinite sample space” or “infinite number of trials”.
- (ii) Many students confused the concept of an empirical distribution tending towards a theoretical distribution (as the number of repetitions in the simulation was increased) with the idea that a population would conform to some theoretical distribution if we were given access to it (i.e. via a census).

It is difficult to distinguish between lack of clarity on terminology (particularly in ii above) and lack of conceptual understanding. This did, however, provide useful information for the teaching staff on the need to guide students more carefully around these subtleties of both language and concept.

In question 6 of the questionnaire we specifically attempted to gauge the didactic effectiveness of the Poisson distribution VBA simulation module. The

simulation considers the number of potholes found in a particular length of road when all that is known is that the average occurrence of potholes per kilometre is some average number. The question attempted to assess whether students grasped the distinction between an empirical and theoretical distribution. It can be seen that the responses in Figures P.6 and P.7 are significantly different from those that would be expected based on guesswork on the students' part (i.e. 50% correct).

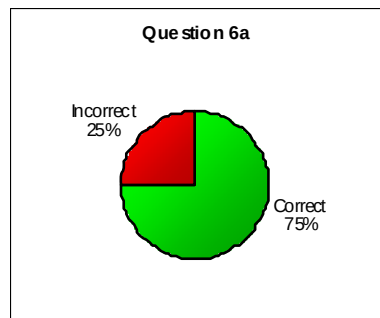


Figure P.5. Question 6a

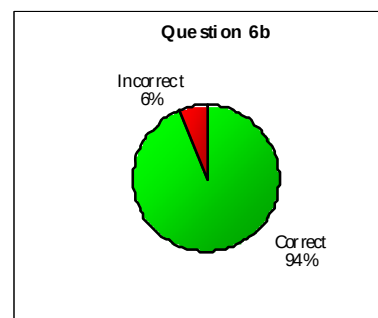


Figure P.6. Question 6b

In excess of 80% of students indicated that the distribution would be Normal, tend to Normal or be approximately Normal in response to question 6c.

7. Conclusions

It is thought that despite the fact that the results demonstrate limitations in students' understanding, the lecturers' assessments of the response of students in the class, and in particular the nature of the questions raised in class, indicate a positive trend. These conclusions are clearly preliminary and need to be supported by an ongoing longitudinal analysis which takes careful cognisance of any potential confounding factors. They are, however, supportive of the

tentative positive conclusions reached by Francis et al. (2007) at Swinburne university. The student body at that university would, of course, constitute a somewhat different socio-economic profile to that of the University of Cape Town. However, the sample of students considered in this analysis was very large (greater than 1000) and it was unanimously felt by the teaching team that exposure to the simulation of data sampling and building of statistical distributions provides a better basis for building an understanding of statistical concepts than that provided by rote application of “statistical recipes”.

In particular, it would appear that the Central Limit Theorem was very effectively taught by simulation and gave students a good chance of grasping a rather subtle idea. It was noted, as expected, that students still exhibited some confusion around the primary concepts of sample size, number of repetitions of an experiment and the idea of an empirical distribution of data and its relationship to an underlying theoretical distribution. However, we remain cognisant of the fact that at least some of this confusion may relate to an inability to express these ideas (that is, not having the words to explain the concepts). This is borne out by the contrast in results displayed in Figure P.3 & P.4 (indicate poor understanding when having to explain in own words) versus Figure P.6 & P.7 (indicate relatively good understanding when identifying features correctly).

Despite the fact that the results indicate there are still the inevitable gaps in conceptual understanding, most students appeared to derive benefit from the new approach and there was evidence of an appreciation of issues relating to, for example, the Central Limit Theorem, which has been absent in previous first year classes. Further evaluation will be done on a regular basis and the teaching material will need to be further refined and improved.

An issue raised by a member of the teaching team was the benefits of using “real-life” data sets as a way of improving the students’ appreciation of the effects of variation and/or error arising from a multitude of sources and how it can best be managed so as to provide information with integrity. There is no doubt that there is considerable value in students’ analysing such “real” data sets with their associated idiosyncrasies and anomalies. However, it must be stressed that this teaching project, in its first phase, is focused on putting across a set of fundamental statistical principles through a teaching approach which uncovers the nature of random variation through repeated sampling. It is posited that “real-life” data sets are not the ideal tool for teaching these principles to beginners as they will throw up a set of questions and attendant complexities which are beyond the grasp of most beginners. It would be envisaged that the second and particularly third year level courses are the more appropriate fora in which to consider applications to “real-life” data.

We believe this approach can form part of a new look at teaching methodology, particularly in the mathematical sciences at Universities. University teaching in South Africa is itself in a state of flux, as it grapples with a very different profile of student and has very different objectives to those of twenty years ago. As universities shift to new approaches that can accommodate those who have had a disadvantaged educational background, university teachers must in turn innovate and look towards methods that work and are better in a fundamental sense as well as appropriate to those who have been historically exposed to flawed and substandard teachers and teaching methods. We believe the approach outlined above is just one such approach.

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Appendix: Evaluation questionnaire:

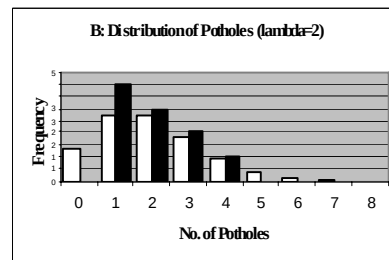
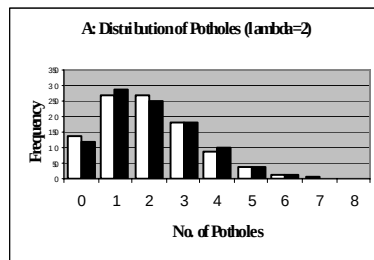
1. Did you enjoy the teaching sessions which used computer simulation?

Yes	
No	
They were OK	

2. Would you say that you understood...

None of the concepts being discussed?	
Some of the concepts being discussed?	
Most of the concepts being discussed?	
All of the concepts being discussed?	

3. What is meant by a probability distribution?
 4. What is meant by an empirical distribution?
 5. When will an empirical distribution of data look exactly like the theoretical distribution (with 100% certainty)?
 6. Consider graphs A and B below which represent frequency histograms for the number of potholes found (per section) in a random sample of 2-km sections selected from 1000 km of road.



- (a) Which colour bars would you expect to represent the theoretical frequencies?

Black bars	
White bars	

- (b) Which of the graphs do you think was generated using a sample size of 10, and which using a sample size of 1000?

Graph A	
Graph B	

- (c) If the experiment using a sample size of 10 was repeated 100 times, and the mean of each sample was calculated, what would you expect to be the shape of the distribution of the sample means?